ST HELENS, Australia

WMO: **959810**

Lat: **41.3381S** Lon: **148.2792E** Elev: **49** StdP: **100.74** Time Zone: **10.00 (AUS)** Period: **01-19** WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

	Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF	
				99.6%			99%			0.4%		1%					
		99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(1)	7	1.8	2.7	-2.3	3.1	7.9	-1.2	3.4	8.0	10.1	11.5	9.2	11.8	3.3	320	0.518	(1)

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

	Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB		
			0.4%		1%		2%		0.4%		1%		2%				
			DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(2)	1	8.1	26.6	17.8	24.2	17.4	22.7	17.0	20.0	23.2	19.2	22.0	18.4	21.2	4.9	310	(2)

	Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB	
	0.4%			1%			2%			0.4%		1%		2%			
	DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(3)	19.0	13.9	21.1	18.2	13.2	20.4	17.4	12.5	19.8	57.5	23.0	55.0	22.2	52.5	21.2	25.5	(3)

Extreme Annual Design Conditions

	Extreme Annual WS				Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
					Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
	1%	2.5%	5%		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
	(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4)	9.6	8.2	7.2	DB	0.0	33.0	0.4	2.4	-0.3	34.7	-0.5	36.1	-0.8	37.5	-1.1	39.2	(4)
(5)				WB	-0.5	21.4	0.5	0.9	-0.9	22.1	-1.1	22.6	-1.4	23.1	-1.8	23.8	(5)

Monthly Climatic Design Conditions

				Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
				(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours		DBAvg	13.2	17.6	17.2	16.3	13.9	11.5	9.7	9.1	9.5	10.8	12.4	14.3	15.9	(6)
(7)			DBStd	3.79	2.74	2.33	2.67	2.42	2.28	2.02	1.88	2.05	2.12	2.53	2.68	2.50	(7)
(8)			HDD10.0	137	0	0	0	1	9	29	41	35	15	5	1	0	(8)
(9)			HDD18.3	1956	47	47	76	135	211	259	287	274	226	186	124	84	(9)
(10)			CDD10.0	1288	234	201	194	118	57	21	13	19	39	79	130	183	(10)
(11)			CDD18.3	63	23	14	12	1	0	0	0	0	0	1	4	8	(11)
(12)			CDH23.3	312	127	48	36	4	0	0	0	0	0	8	39	50	(12)
(13)			CDH26.7	82	43	11	6	0	0	0	0	0	0	1	9	12	(13)
(14)	Wind		WSAvg	3.6	4.0	3.5	3.4	3.4	3.4	3.4	3.4	3.6	3.6	3.8	3.9	4.0	(14)
(15)	Precipitation		PrecAvg	747	60	43	62	66	64	56	70	66	62	58	69	86	(15)
(16)			PrecMax	1135	270	186	177	250	197	167	157	198	184	170	189	392	(16)
(17)			PrecMin	458	4	1	13	2	15	15	6	1	11	2	3	4	(17)
(18)			PrecStd	176	56	40	41	59	46	33	42	45	41	39	45	80	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB		31.9	28.5	26.8	23.5	19.3	16.6	15.5	17.2	19.9	24.4	28.0	28.3	(19)
(20)			MCWB		19.4	19.1	18.6	16.6	14.2	12.5	11.4	11.7	12.9	14.4	17.8	18.2	(20)
(21)		2%	DB		26.6	24.3	23.8	21.1	17.7	15.4	14.4	15.6	17.8	20.7	23.2	24.2	(21)
(22)			MCWB		17.7	18.3	18.3	15.8	13.6	12.1	10.7	11.1	12.1	13.7	15.7	16.8	(22)
(23)		5%	DB		23.8	22.6	22.0	19.4	16.7	14.5	13.6	14.5	16.3	18.5	20.7	22.1	(23)
(24)			MCWB		17.2	17.8	17.4	15.3	13.2	11.5	10.3	10.6	11.5	13.1	15.0	16.3	(24)
(25)		10%	DB		22.1	21.3	20.6	18.1	15.9	13.7	12.8	13.5	15.0	17.0	19.0	20.5	(25)
(26)			MCWB		17.2	17.3	16.9	14.8	12.7	10.9	9.9	10.2	11.0	12.5	14.5	15.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB		21.2	20.8	20.6	18.2	15.8	14.2	12.5	13.0	14.4	16.1	18.7	19.6	(27)
(28)			MCDB		25.8	24.4	23.5	20.6	17.0	15.3	13.9	14.6	16.8	18.7	23.1	24.1	(28)
(29)		2%	WB		19.8	19.9	19.6	17.2	14.7	12.8	11.6	12.1	13.3	15.1	17.4	18.5	(29)
(30)			MCDB		23.8	22.4	22.1	19.3	16.8	14.4	13.4	14.3	16.1	17.8	21.5	21.8	(30)
(31)		5%	WB		18.9	19.1	18.6	16.3	13.9	12.0	11.0	11.3	12.4	14.2	16.2	17.5	(31)
(32)			MCDB		21.8	21.4	21.1	18.6	16.3	13.7	13.0	13.8	15.3	17.3	19.4	20.4	(32)
(33)		10%	WB		18.0	18.2	17.6	15.2	13.0	11.1	10.3	10.6	11.6	13.3	15.1	16.5	(33)
(34)			MCDB		21.1	20.6	20.1	17.8	15.4	13.2	12.5	13.2	14.4	16.7	18.0	19.4	(34)
(35)	Mean Daily Temperature Range		MDBR		8.1	7.2	7.7	7.7	7.6	7.2	7.3	7.9	8.4	8.2	8.2	8.0	(35)
(36)		5% DB	MCDBR		11.7	9.6	9.7	9.4	8.1	7.8	7.6	8.6	10.1	10.5	11.4	10.9	(36)
(37)			MCWBR		5.3	5.1	5.2	5.3	5.2	5.1	5.0	5.3	5.8	5.6	5.6	5.4	(37)
(38)		5% WB	MCDBR		9.6	7.5	8.1	8.0	6.8	6.7	6.6	7.7	8.4	8.7	9.4	8.8	(38)
(39)			MCWBR		5.2	4.9	4.9	5.1	5.0	4.7	4.8	5.2	5.5	5.4	5.4	5.1	(39)
(40)	Clear-Sky Solar Irradiance	taub			0.351	0.346	0.336	0.321	0.305	0.296	0.294	0.310	0.327	0.338	0.345	0.344	(40)
(41)		taud			2.516	2.546	2.567	2.586	2.599	2.610	2.628	2.576	2.524	2.500	2.510	2.522	(41)
(42)		Ebn at Noon			976	955	921	868	816	791	823	867	915	952	977	989	(42)
(43)		Edh at Noon			110	101	91	77	64	58	61	76	93	105	110	110	(43)
(44)	All-Sky Solar Radiation	RadAvg			6.65	5.60	4.48	3.06	2.06	1.62	1.84	2.68	3.96	5.20	6.05	6.83	(44)
(45)		RadStd			0.45	0.30	0.21	0.17	0.11	0.13	0.08	0.19	0.28	0.25	0.42	0.41	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (22 neighbors)	+0.33	N/A	N/A	+0.69	+0.34	N/A	N/A	-102	+109	+18	(47)

Nomenclature: See separate page